Response Surface Analysis of ERCOT Day-Ahead–Real-Time Price Spreads

The Role of Temperature, Electricity Demand, and Natural Gas Prices

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Summary

Using hourly observations from ERCOT's four competitive load zones in 2025, this study examines nonlinear relationships between temperature, electricity demand, natural gas prices, and day-ahead–real-time (DA–RT) electricity price spreads. The quadratic specification explains little of the variation in signed DA–RT spreads, but second-order response surface regressions reveal statistically significant curvature in spread magnitude. However, the additional nonlinear terms do not consistently improve performance in later months. Extreme spreads also occur substantially more frequently during periods of high system net load, about 13 times as often as in other hours. These findings highlight both the value and the limitations of using realised market fundamentals to characterise electricity price risk. All results explain past hours using conditions known only after the fact; none is a real-time forecast.

1. Introduction

This note describes how temperature, electricity demand and natural gas prices relate to the gap between day-ahead and real-time electricity prices in ERCOT's load zones. It follows the steps of the assignment: collecting the data, calculating the required measures, applying response surface regression, and reporting the findings.

A response surface regression (RSM) approximates an outcome with a low-order polynomial in several inputs (NCSS Chapter 330; Myers et al. 2016). With temperature T, zonal load L and gas price G, each rescaled to a coded value x, the full second-order surface is

$$\begin{aligned} y = {} & \beta_0 + \beta_T x_T + \beta_L x_L + \beta_G x_G \\ & + \beta_{TT} {x_T}^2 + \beta_{LL} {x_L}^2 + \beta_{GG} {x_G}^2 \\ & + \beta_{TL} x_T x_L + \beta_{TG} x_T x_G + \beta_{LG} x_L x_G + \varepsilon. \end{aligned}$$

The squared terms let a relationship bend: a T^2 term lets load fall as weather moves from cold to mild and rise again as it turns hot. The interactions let the effect of one input depend on another: a T × L term lets the spread respond more strongly to temperature when load is already high. Because the model is linear in its coefficients, ordinary least squares (OLS) estimates it. Canonical analysis then classifies the fitted quadratic as a bowl, a dome, a saddle or a ridge.

Two findings from prior work motivate the analysis. Electricity demand is a nonlinear function of temperature (Engle et al. 1986), which motivates second-order terms and raises the question of whether that curvature appears in each zone. The premium of forward over spot electricity prices depends on the variance and skewness of spot prices (Bessembinder & Lemmon 2002), which motivates studying the size of the DA–RT spread as well as its sign. ERCOT's load zones also differ in their mix of weather-driven and industrial demand, so each zone is modelled separately. The research question is how temperature, demand and gas jointly relate to zonal DA–RT spreads and implied heat rates, and what allowing that relationship to curve adds.

2. Data

The sample covers the 2025 operating year (1 January–31 December) for the Houston, North, South and West load zones: 8,760 hours per zone and 35,040 zone-hours. The 2024 year is processed through the same pipeline as a partial replication: prices, load, weather and gas are complete for all 8,784 hours, but ERCOT's forecasts of available wind and solar capacity are complete only from 24 September 2024, so analyses that use forecast surprises are not replicated. Table 1 lists the sources.

All series are joined on UTC timestamps, which preserves the 23- and 25-hour daylight-saving days. No zone-hour is missing or duplicated, and none is trimmed. The load total correlates at 0.997 with ERCOT's native-load archive, spot checks of two stations (Houston Intercontinental in July, Dallas–Fort Worth in January) agree with NOAA records to within 0.08 °F, and the gas series matches FRED on every date.

Prices are ERCOT's final settled values from its annual archives. Grid Status's first-published series serves as a cross-check: every one of the 35,040 DA hours and 140,160 RT intervals matches by key, but values differ for 89 DA hours on two dates (27 June and 6 December) and 135 RT intervals on eight dates. The largest difference is \$1,595.73/MWh: an RT interval on 30 June that appears at \$1,624–\$1,636/MWh in the first-published data but settled at \$28–\$40. Where the two differ, the settled value is used.

Several inputs are proxies rather than exact matches. Load is reported by ERCOT forecast zone, which includes the publicly owned and cooperative utilities that settle at their own load zones (Austin Energy, CPS Energy and LCRA within South; Rayburn within North), so load covers a larger footprint than the matching load-zone price. Henry Hub is a Louisiana benchmark, not a delivered Texas fuel cost. Weighted averages of two to six airport stations are coarse stand-ins for mesoscale temperature fields.

Table 1. Raw series.

Series	Native resolution	Source	Hourly treatment
DA price, \$/MWh	1 hour	ERCOT settled DAM prices (NP4-180-ER)	Load-zone settlement point prices (LZ_HOUSTON, LZ_NORTH, LZ_SOUTH, LZ_WEST).
RT price, \$/MWh	15 minutes	ERCOT settled RTM prices (NP6-785-ER)	Mean of the hour's four intervals; all four present in every hour.
Load, MW	1 hour	ERCOT actual load by forecast zone, via Grid Status	Measured demand (not a forecast).
Wind, solar, MW	1 hour	ERCOT fuel mix; wind and solar actuals and forecasts, via Grid Status	Generation, available capacity, and the latest forecasts published before 09:45 Central on the previous day (the 09:30 load and 08:55 wind and solar reports).
Temperature, °F	∼hourly	NWS/FAA ASOS, Iowa Environmental Mesonet	Fixed-weight average of 2–6 airport stations per zone (below).
Gas, \$/MMBtu	daily	EIA Henry Hub spot (RNGWHHD)	Last trade date before the operating day.

Station weights (%, judgement calls approximating metro population): Houston IAH 50, HOU 50; North DFW 50, DAL 40, ACT 10; South SAT 38, AUS 33, MFE 13, CRP 6, BRO 6, LRD 4; West MAF 35, ODO 35, ABI 18, SJT 12. Gaps of up to 3 hours are interpolated.

3. Required measures

The *spread* is $S = P_{\mathrm{DA}} - P_{\mathrm{RT}}$ for the same delivery hour, so a positive value is a DA premium; a DA price of \$40 and an RT price of \$50 give $S = -\$10$/MWh.

The *implied market heat rate* is IMHR $= P/G$ in MMBtu/MWh, computed for DA and RT; \$40/MWh power with \$4/MMBtu gas gives 10. It is a price ratio, not a measured plant efficiency, and it is negative when power prices are.

The *load factor* is mean load divided by each zone's own annual peak.

Table 2 summarises the zones:

- **Load.** Houston, North and South are weather-driven: a quadratic in temperature explains 85–89% of their hourly load. West's load is flatter and much less weather-sensitive (R^2 0.32, load factor 0.80).
- **Spreads.** The typical hour carries a small DA premium (median +$1.9 to +$2.8; DA above RT in 59–63% of hours). Rare, large RT spikes pull the mean spreads to about zero.
- **Heat rates.** In the eastern zones the DA heat rate is 2.5–2.9 MMBtu/MWh higher on-peak than off-peak.

Table 2. Zonal load measures, representative DA–RT spreads and implied heat rates, 2025.

2025, 8,760 hours per zone	Houston	North	South	West
Load				
Mean load (GW)	14.0	18.4	14.6	9.5
Peak load (GW)	21.7	32.4	23.2	11.9
Load factor (mean / peak)	0.65	0.57	0.63	0.80
R^2 of load on T + T^2	0.89	0.85	0.86	0.32
DA − RT spread ($/MWh)				
Mean	+1.4	+0.7	−0.2	+0.5
Median	+2.1	+2.2	+1.9	+2.8
On-peak mean	+1.8	+0.7	−0.3	+1.2
Off-peak mean	+1.0	+0.8	−0.1	−0.1
Mean |spread|	10.4	11.6	11.3	15.9
1st / 99th percentile	−63 / +49	−70 / +55	−73 / +43	−83 / +65
Hours with DA > RT	63%	62%	61%	59%
Implied market heat rate (MMBtu/MWh)				
Mean DA / mean RT	10.1 / 9.7	9.7 / 9.4	9.7 / 9.6	12.4 / 12.3
DA on-peak / off-peak	11.7 / 8.8	11.1 / 8.5	11.0 / 8.6	12.3 / 12.5

Hours are equally weighted. On-peak means weekdays, hours beginning 06:00–21:00 Central, excluding NERC holidays.

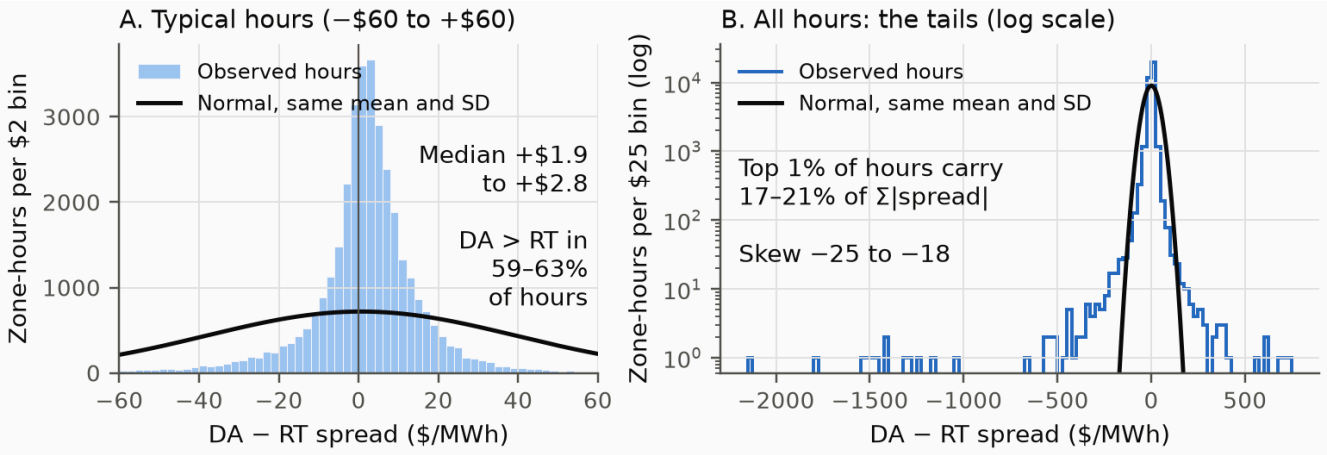


Figure 1. Hourly DA–RT spread, all zones pooled, against a normal curve with the same mean and standard deviation. The spread is near zero in most hours but has extreme tails, mostly from RT spikes: a normal distribution would expect 0.4 zone-hours beyond ±$170, whereas 144 lie there.

4. Exploratory analysis

Figure 2 shows why a quadratic is useful. A straight line misses the V-shape: cold and hot hours both carry more load than mild hours. Adding temperature squared raises R^2 from 0.12–0.46 to 0.85–0.89 in the eastern zones. The gain carries forward: in the same month-by-month test used for the price models, the squared term cuts later-month load error by 84–86% in Houston, North and South, and by 25% in West. Because load depends strongly on temperature, the two factors overlap heavily when used together. In Houston, North and South their correlation is 0.70–0.94 in absolute value within each branch of the V (below and above 65 °F); in West it is only 0.31–0.54. The dispersion of the spread is U-shaped in temperature. No zone has a single top-1% $|S|$ hour between 55 and 65 °F. 75% of extreme hours fall in the morning and evening ramps, and all of them fall on 56 days.

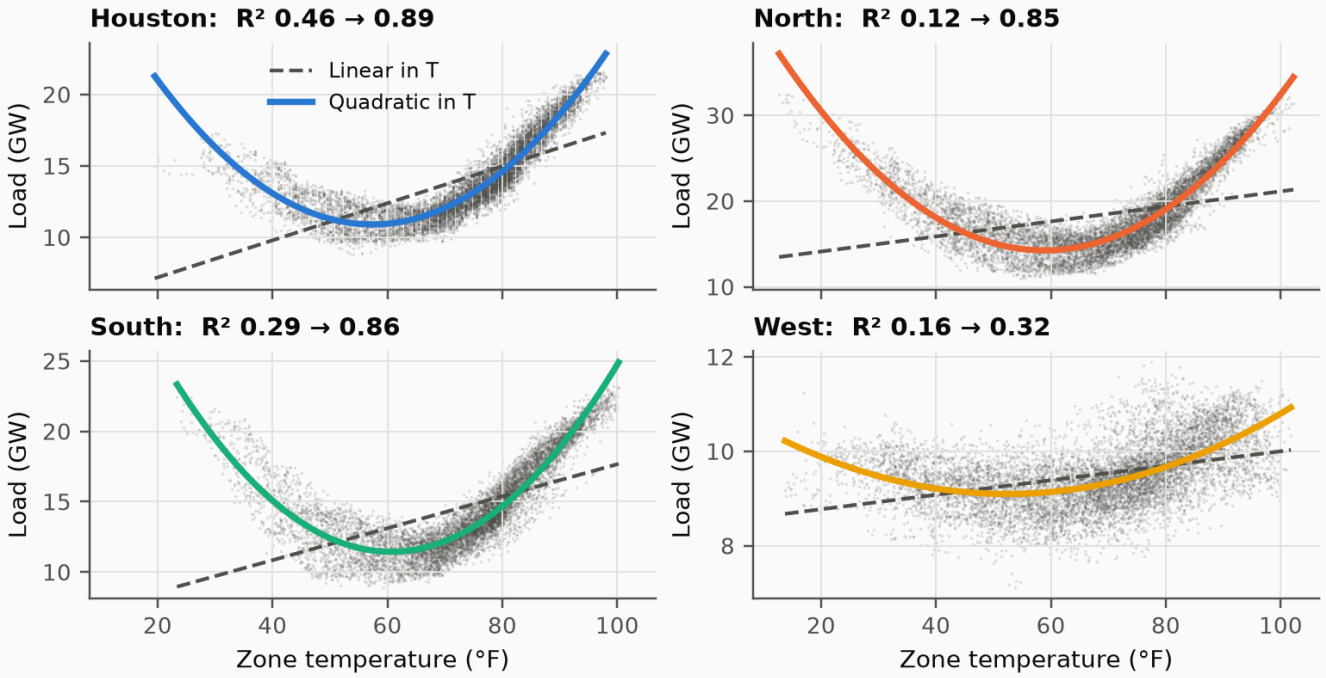


Figure 2. Hourly zonal temperature against zonal load, 2025. Dashed lines fit load on temperature alone; solid lines add temperature squared. The titles give the in-sample R^2 for the linear and the quadratic fit. These descriptive fits are separate from the three-factor price models.

4.1 What makes an extreme hour different

System tightness is measured by *system net load*, in two variants:

- *Realised net load* is load minus wind and solar *generation*: what conventional plants and storage had to serve.
- For forecast *surprises*, load minus wind and solar *available capacity* is compared with the latest forecasts published before 09:45 on the previous day, which reduces the influence of curtailment on the measured surprise.

To ask how much more often large DA–RT divergences occur when the system is tight, I rank hours by realised system net load and compare the tightest 10% with the rest. An hour is called extreme when its $|S|$ is in the zone's own top 1%, so the base rate is 1% by construction and the comparisons are relative risks.

- **Headline.** Extremes occur in 6.0% of the tightest hours, against 0.45% of the others: about 13 times as often, or 11 times (7–18) without the 19–20 February storm. The 90% interval is 8–21×, from a boot-

strap that resamples whole days, because extremes cluster on a few days. The tightest decile contains 53%–68% of each zone's extreme hours.

- **Time of day.** Tight hours are mostly summer evenings, so I repeat the comparison with net load ranked within each hour of day. The ratio roughly halves, to 7.4 (4.2–12.5). The concentration is therefore not explained by time of day alone. Season, weather, renewable output and outages are not held fixed, so this remains a descriptive association, not a causal effect of net load.
- **In dollars.** $|S|$ exceeded \$100/MWh in 5.3% of the tightest hours against 0.39% of the others. RT exceeded DA by more than \$100/MWh, an incremental cost of more than \$100 for each megawatt-hour bought in real time rather than day-ahead, in 2.5% against 0.35%.
- **Replication.** Using the same full-year ranking of realised net load in 2024, the ratio is 12 (7–19).

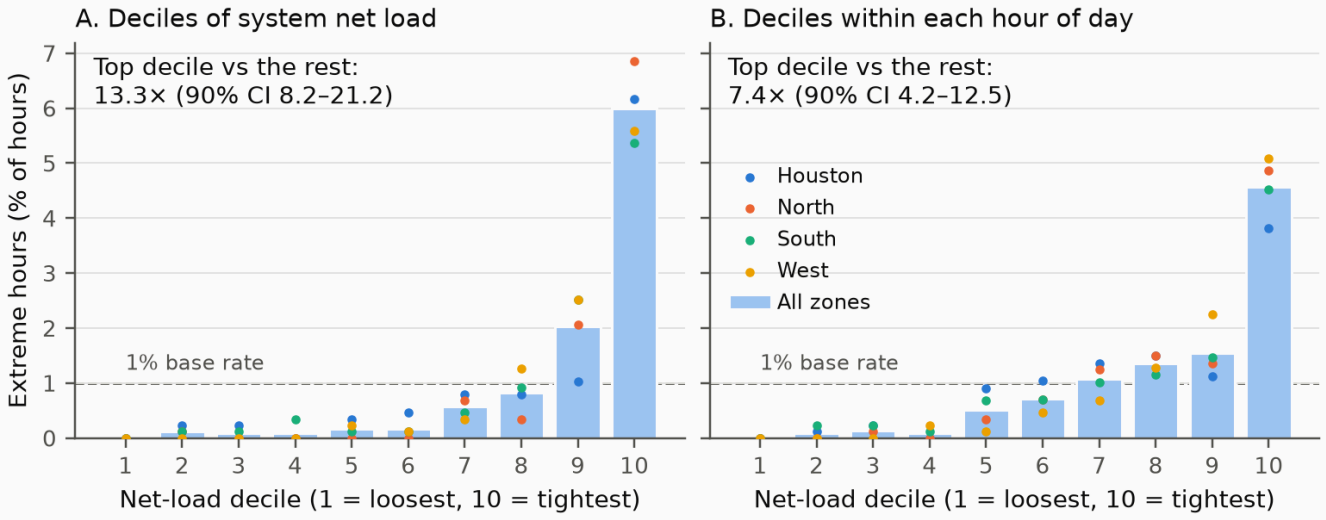


Figure 3. Share of hours with |DA − RT| in the zone's own top 1%, by decile of realised system net load (load minus wind and solar generation). Bars pool the zones, dots show individual zones, and the dashed line is the 1% base rate. Panel A ranks all hours. Panel B ranks within each hour of day, a time-of-day robustness check that does not hold season, weather or renewables fixed. Printed ratios compare the tightest decile with the rest, with 90% day-block bootstrap intervals.

16 May 2025 was retrospectively selected as an illustrative stress event, using a reproducible event-selection rule: the day with the most RT-above-DA extreme hours in the tightest decile, outside the February storm. Figure 4 shows it.

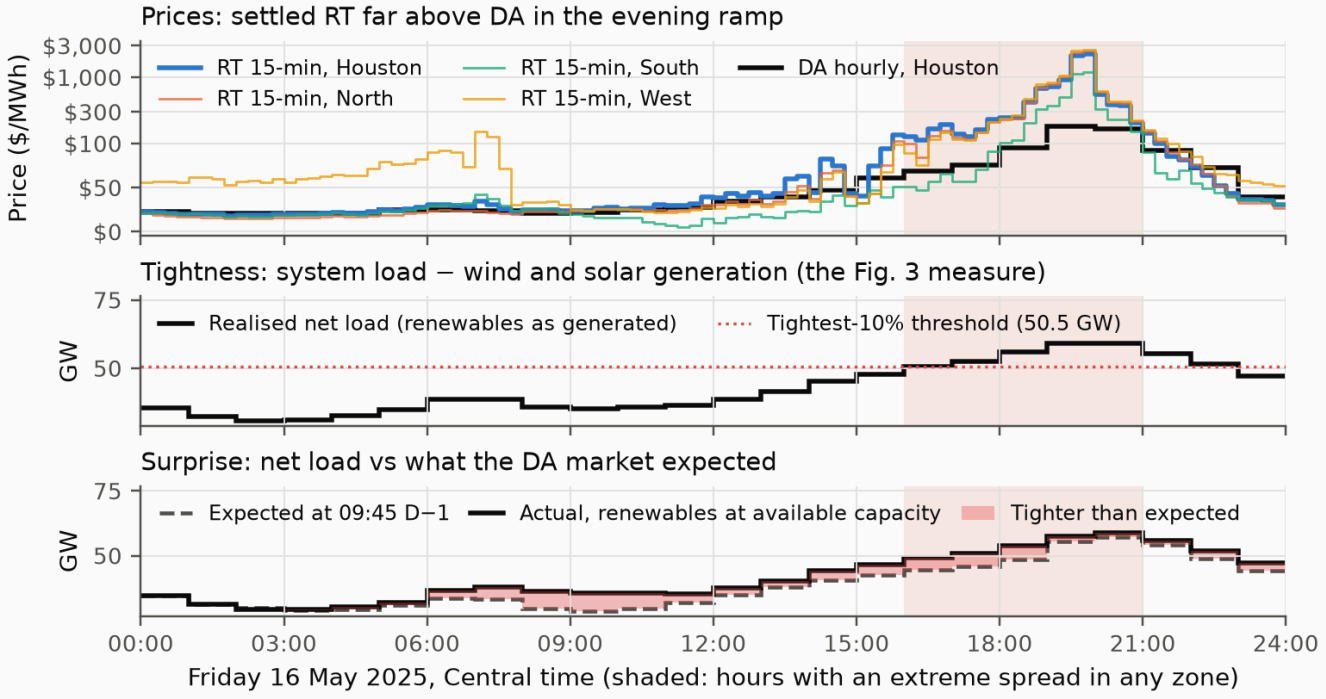


Figure 4. 16 May 2025. Top: settled 15-minute RT prices by zone and Houston's hourly DA price (symmetric-log scale). Middle: realised net load (renewables as generated, the Figure 3 measure) against the tightest-10% threshold. Bottom: net load expected at 09:45 the day before, against actual net load with renewables at available capacity; red marks hours that were tighter than expected. Shaded columns mark hours with an extreme spread in any zone.

- **Tightness.** Realised net load crossed the tightest-10% threshold of 50.5 GW at 16:00, peaked at 59 GW, and stayed above the threshold until 23:00.
- **Surprise.** From the hour beginning 03:00 onward the system ran tighter than the forecast the DA market had cleared on. In the extreme hours the gap was 1.6–5.2 GW, with available wind 2.9–3.7 GW below forecast.
- **Prices.** Houston's hourly RT price reached $1,507/MWh in the hour beginning 19:00, against a DA price of $184, and one 15-minute West interval (beginning 19:45) settled at $2,621. Houston had extreme hours from the hour beginning 16:00, and all four zones did in the hours beginning 18:00 to 20:00 (14 zone-hours in total).

The day also shows the limits of the explanation. The largest surprise was in the hour beginning 09:00, when the system was 6.7 GW tighter than expected, yet the Houston spread was only about −$4. In the hour beginning 19:00 the surprise was a smaller 1.8 GW (expected 55.6 GW, actual 57.4 GW), with expected net load close to the day's maximum of 57.2 GW in the following hour, and the Houston spread reached −$1,324. For that hour, the surprise surface (a quadratic in the net-load surprise and expected net load, fitted to Houston hours from January–April 2025 and given the realised surprise) predicts −$1.90; the core quadratic in realised T, L and G predicts +$9.81, the wrong sign. Neither prediction comes close to the observed size. Realised net load also stayed above the tightest-10% threshold for two hours after the extremes ended. A single day illustrates the pattern; it does not prove it.

5. Response surface regression

Each zone gets separate models for three responses: the signed spread S, which describes direction; its magnitude $|S|$, which describes the size of the mismatch and is a risk proxy rather than a conditional variance; and the DA implied heat rate.

The models form a nested ladder, so that each comparison isolates one change:

- **M0**, the benchmark, predicts each hour by the training months' average for that hour of day;
- **M1** is linear in T, L and G;
- **M2** adds the three squares and three interactions (the RSM);
- **M1H** and **M3** add hour-of-day indicators to M1 and M2, so that the daily cycle is not credited to curvature.

The regression models (M1, M2, M1H and M3), but not the M0 benchmark, include an intercept shift for the 27 days after ERCOT's 5 December 2025 market redesign, Real-Time Co-optimization Plus Batteries (ERCOT 2025). They are estimated by OLS on factors coded so that their 1st and 99th percentiles map to ± 1. The implementation reproduces NCSS's worked example to every printed digit.

Because the data are observational rather than a designed experiment, three departures from NCSS are needed:

- **Collinearity.** Variance inflation factors reach 51, so I interpret fitted surfaces rather than individual coefficients, and draw surfaces only where comparable hours exist.
- **Dependence.** Hourly temperature and load are strongly serially dependent. Their effective sample size for estimating a mean, computed from autocorrelations up to one week (168 lags) with Bartlett weights, is only 70–79 for temperature and 102–124 for load in each zone, out of 8,760 hours. Inference therefore uses Newey–West (HAC) standard errors with 48 lags. Across zones and responses, the median ratio of HAC to classical standard errors ranges from 1.9 to 6.5 (the largest single ratio is 8.3). Curvature is judged by joint Wald tests.
- **Validation.** Each model, and the M0 benchmark, is refitted on all months before a test month and evaluated on that month: January–March predict April, January–April predict May, and so on through December. The test window is therefore April–December, 6,601 hours per zone. The 19–20 February cold snap is always in training and never in a test month, so these errors exclude it. Skill is $1 - \mathrm{SSE}/\mathrm{SSE}_{\mathrm{M0}}$; positive values beat the hourly benchmark.

These are explanatory models, not forecasting models. Every model uses the realised temperature, load and gas of the hour it describes, none of which is known when DA bids close at 10:00 the day before. The forward test therefore checks whether a relationship is stable over time; it is not a trading backtest, which would require bid-time information, a bidding rule and transaction costs.

6. Results

Because this is a response-surface assignment, the central question is what the quadratic and interaction terms accomplish. The fair comparison keeps the inputs, hours and controls unchanged and adds only those six terms. Adding terms to an OLS regression can never lower its in-sample R^2, so a higher R^2 alone does not show that curvature is useful. R^2 measures variation explained in the fitting sample, not prediction accuracy. The real test is whether the extra flexibility lowers the error in months the model has not seen. Table 3 reports both.

Signed spread. Curvature does not help. R^2 rises only from 0.2–1.8% to 0.5–4.9%, the six terms are jointly significant only in West, and later-month error *increases* in every zone (+0.3% to +5.2%). Temperature, load and gas levels explain little of the hourly DA–RT gap under this specification. The in-sample improvements did not generalise to later months, consistent with overfitting or unstable relationships. This is a negative result, and it is reported as one.

Table 3. Linear and quadratic models, 2025. Each pair differs only by the six second-order terms (M1 against M2; with hour-of-day indicators, M1H against M3). "Later-month error" is the change in pooled squared error over April–December when the six terms are added: positive means the quadratic model is worse. "Mild" and "hottest 1%" are fitted values at the mean conditions of the 60–70 °F band and of each zone's hottest 1% of hours. No hours or extreme prices are removed.

2025, per zone	Houston	North	South	West
Signed spread				
R^2, linear → quadratic	0.002 → 0.005	0.006 → 0.009	0.017 → 0.031	0.018 → 0.049
Joint test of the six terms, HAC p	0.358	0.077	0.228	<0.001
Later-month error, quadratic vs linear	+0.3%	+0.3%	+5.2%	+3.7%
Spread magnitude |spread|				
R^2, linear → quadratic	0.027 → 0.035	0.034 → 0.062	0.048 → 0.088	0.032 → 0.114
Joint test of the six terms, HAC p	0.020	0.033	0.004	<0.001
Linear fit, mild → hottest 1% ($/MWh)	6.5 → 19.0	6.1 → 21.7	6.6 → 18.7	18.9 → 9.1
Quadratic fit, mild → hottest 1% ($/MWh)	7.9 → 20.6	8.6 → 26.0	7.5 → 22.7	10.8 → 34.6
Later-month error, quadratic vs linear	+2.2%	+6.3%	+2.8%	+0.5%
DA implied heat rate				
R^2, linear → quadratic	0.18 → 0.18	0.14 → 0.15	0.20 → 0.21	0.07 → 0.21
R^2, with hour of day	0.37 → 0.38	0.34 → 0.36	0.36 → 0.37	0.33 → 0.40
Later-month error, with hour of day	+3.9%	+3.5%	+7.7%	−12.0%

Spread magnitude. Here the six terms are statistically significant in sample, but they do not lower later-month error:

- **In sample.** The six terms are jointly significant in every zone (HAC $p \leq 0.033$), and R^2 rises from 2.7–4.8% to 3.5–11.4%.
- **Shape.** The linear model in T, L and G already predicts larger magnitudes at both temperature extremes in Houston, North and South, through the load term, since load is V-shaped in temperature. In West, however, it places the hottest hours below mild ones ($9.1 against $18.9), while the quadratic gives $10.8 against $34.6. Quantile regressions on the same quadratic design, estimating the 5th and 95th percentiles of the spread, show the band between them widening 1.9–3.6-fold from mild to the hottest hours.
- **Later months.** Squared error is still higher than the linear model's in every zone (+0.5% to +6.3%). The in-sample curvature did not translate into lower error in later months.

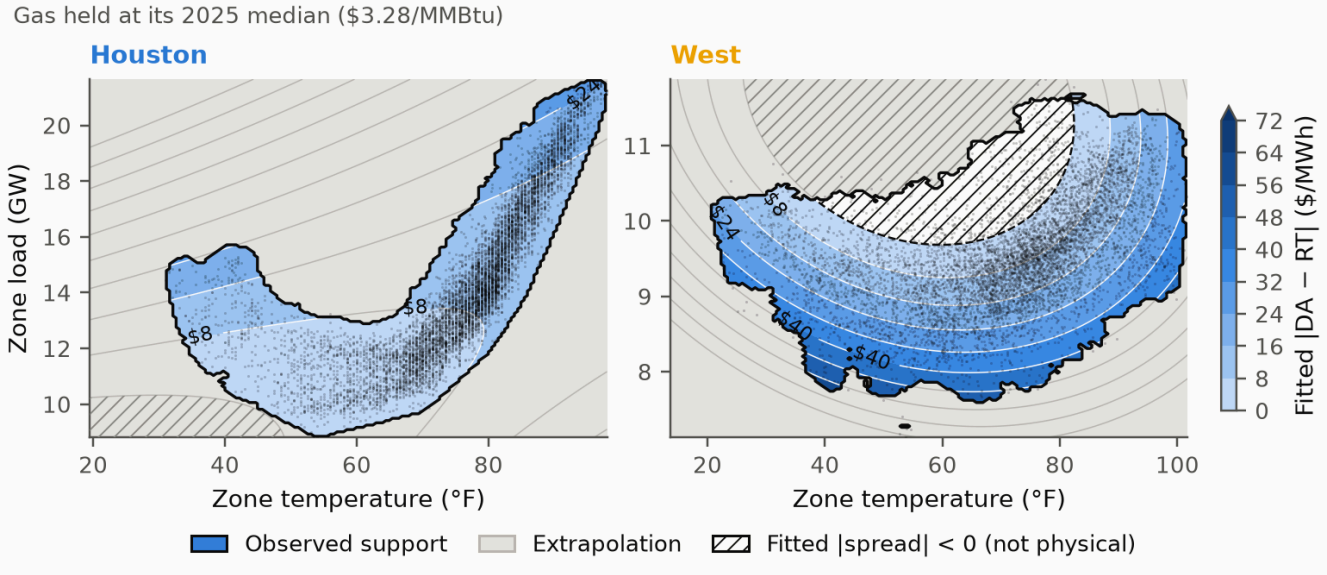


Figure 5. Fitted |DA − RT| from M2 over temperature and zonal load, at median gas. Coloured, outlined regions are observed support, where hours with near-median gas (dots) lie close by. Grey regions are extrapolation. Hatched regions are fitted magnitudes below zero, which are impossible (6.3% of West's fitted hours; South 2.8%). The canonical analysis gives saddles and a ridge, so no optimum is read from the stationary point.

In Houston, fitted magnitude rises along the hot, high-load arm of the V. In West it is highest at low load at every temperature: a joint pattern that no single marginal effect shows. The hatched area is the cost of an unconstrained quadratic for a non-negative outcome, so the figure shows where magnitude rises, not its level near zero.

DA heat rate. Most of the model's success comes from terms other than curvature:

- **Without an hour control,** curvature adds almost nothing in the eastern zones (R^2 +0.002 to +0.011). In West it nearly triples R^2 (0.07 to 0.21).
- **With identical hour-of-day controls,** adding curvature increases later-month error in the eastern zones (+3.5% to +7.7%) and reduces it only in West (−12.0%). The heat-rate model's overall fit should not be attributed to RSM curvature.

The realised surface also misses what Section 4.1 suggests matters for direction. Against the 09:45 forecasts, the net-load surprise has a rank correlation of −0.35 to −0.47 with the spread, compared with −0.01 to +0.09 for realised temperature and load. On 16 May, however, a surprise model got the sign of the 19:00 spread right but missed its size by about $1,322/MWh.

7. Conclusions

1. **The measures.** A transparent hourly panel of settled prices provides the required zonal spreads, load measures and implied heat rates.
2. **Curvature.** It is clear in demand and in the size of price risk, but not in its direction. For spread magnitude, the six quadratic terms are jointly significant in every zone, raise in-sample R^2, and change the fitted shape (in West, the quadratic ranks the hottest hours above mild ones, the linear fit below). It adds nothing for the signed spread and does not improve predictions for later months. Why the signed spread is not explained by realised conditions is not tested here.
3. **Tight hours.** Extreme spreads concentrate in them: about 13 times as often, and 7.4 times when ranking within hour of day. RT spikes coincide with the system running tighter than expected, as on 16 May 2025.
4. **Zones.** What curvature buys depends on the zone: little for the eastern heat rates, a great deal in West.

The limitations matter for how far the results can be pushed:

- **One year.** The patterns describe 2025. In the full-year 2024 data, the V-shaped load, the curvature of spread magnitude and the concentration of extremes in tight hours recur, but other results do not. For example, the quadratic heat-rate model with hour-of-day controls beat the hourly benchmark in later months of 2025 (skill $+0.09$ to $+0.13$) but not in three of four zones in 2024 (-0.19 to $+0.01$). The surprise results could not be checked in 2024 at all.
- **Association, not cause.** The regressions describe how prices moved with conditions; they do not show that the conditions caused the moves. Load may also respond to price, for example when flexible loads curtail during spikes, and the tightness comparison does not hold season or weather fixed.
- **Explanation, not forecasting.** Every model uses conditions known only after the fact. Even with that information, on 16 May the surprise model matched the sign of the largest spike but missed its size by about \$1,322/MWh.
- **Measurement.** Station-weighted temperatures, a national gas price and imperfectly matched zone boundaries are proxies. An unconstrained quadratic can also predict impossible negative magnitudes.

Natural next steps are gridded (mesoscale) zonal temperature, weather-forecast errors and a regional gas price. If the goal becomes explaining spikes, reserve-margin and outage data would be the targeted additions; more polynomial terms alone have not solved the problem.

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